

## 1.2 Classwork Solutions: Speed, Velocity, and Unit Conversion

*Notation:*  $=$  joins values that are exactly equal;  $\approx$  introduces a rounded value. Students should write it the same way — once you round the calculator display, the equals sign is no longer true. Each conversion runs as one chain of unit factors, so there is nothing to round in the middle.

1. A student walks 24.0 m in 18.5 s.

(a)  $v = \frac{d}{t} = \frac{24.0 \text{ m}}{18.5 \text{ s}} = 1.2972 \dots \approx \mathbf{1.30 \text{ m/s}}$

- (b) **Three significant figures.** Both measurements have three, and a quotient carries no more than the least precise measurement that went into it. Writing 1.29729 would claim a precision the stopwatch and the tape measure do not have.

2. Convert each speed.

(a)  $30.0 \frac{\text{km}}{\text{h}} \times \frac{1000 \text{ m}}{1 \text{ km}} \times \frac{1 \text{ h}}{3600 \text{ s}} = \frac{30,000}{3600} \frac{\text{m}}{\text{s}} = 8.333 \dots \approx \mathbf{8.33 \text{ m/s}}$

*The km cancel and the h cancel, leaving m/s. A useful benchmark: dividing km/h by 3.6 gives m/s.*

(b)  $25.0 \frac{\text{m}}{\text{s}} \times \frac{1 \text{ km}}{1000 \text{ m}} \times \frac{3600 \text{ s}}{1 \text{ h}} = \mathbf{90.0 \text{ km/h}}$  (exact — nothing to round)

3. A fastball at 95 mi/h.

(a)  $95 \frac{\text{mi}}{\text{h}} \times \frac{5280 \text{ ft}}{1 \text{ mi}} \times \frac{1 \text{ h}}{3600 \text{ s}} = \frac{501,600}{3600} \frac{\text{ft}}{\text{s}} = 139.33 \dots \approx \mathbf{139 \text{ ft/s}}$

- (b) Chain all the way from the original figure in one expression:

$$95 \frac{\text{mi}}{\text{h}} \times \frac{5280 \text{ ft}}{1 \text{ mi}} \times \frac{1 \text{ m}}{3.28 \text{ ft}} \times \frac{1 \text{ h}}{3600 \text{ s}} = \frac{95 \times 5280}{3.28 \times 3600} \frac{\text{m}}{\text{s}} = 42.48 \dots \approx \mathbf{42.5 \text{ m/s}}$$

*Write the whole conversion as one product and evaluate once. If you continue from part (a) instead, carry the full 139.33 ft/s — restarting from the rounded 139 gives 42.4, wrong in the last figure. Rounding mid-chain is the commonest way to lose an accuracy mark.*

4. Put all three in m/s before comparing.

• Car:  $60 \frac{\text{mi}}{\text{h}} \times \frac{5280 \text{ ft}}{1 \text{ mi}} \times \frac{1 \text{ m}}{3.28 \text{ ft}} \times \frac{1 \text{ h}}{3600 \text{ s}} = \frac{60 \times 5280}{3.28 \times 3600} \frac{\text{m}}{\text{s}} = 26.829 \dots \approx \mathbf{26.8 \text{ m/s}}$

• Train:  $95 \frac{\text{km}}{\text{h}} \times \frac{1000 \text{ m}}{1 \text{ km}} \times \frac{1 \text{ h}}{3600 \text{ s}} = 26.39 \dots \approx \mathbf{26.4 \text{ m/s}}$

• Motorcycle:  $\mathbf{27 \text{ m/s}}$  (already in m/s)

At three figures the order reads train < car < motorcycle — **but that ordering is not supported by the data.**

*Every speed here is given to two significant figures at best — and “60 mi/h” may be*

only one, since the trailing zero is ambiguous. At two figures the three speeds are 27, 26 and 27 m/s, so the defensible answer is: **the train is the slowest, and the car and the motorcycle cannot be told apart.** The 0.2 m/s between 26.8 and 27 is smaller than the uncertainty already in the data; using the third figure to break that tie claims precision the problem never gave. Give full credit for a correctly converted list, and full credit again — the real lesson — for saying the ranking cannot be completed.

5. Sound travels 343 m/s; the delay is 4.0 s.

$$d = vt = (343 \text{ m/s})(4.0 \text{ s}) = 1372 \text{ m} \approx \mathbf{1.4 \times 10^3 \text{ m}} \quad (2 \text{ s.f., because } 4.0 \text{ s has two})$$

$$1372 \text{ m} \times \frac{3.28 \text{ ft}}{1 \text{ m}} \times \frac{1 \text{ mi}}{5280 \text{ ft}} = 0.8523 \dots \approx \mathbf{0.85 \text{ mi}}$$

This is the “count the seconds” rule of thumb: roughly 1 km every 3 seconds, or about 1 mile every 5 seconds.

6. Climbing 5 floors, 18 steps each, 8.0 in per step.

$$(a) \quad 5 \text{ floors} \times \frac{18 \text{ steps}}{1 \text{ floor}} = \mathbf{90 \text{ steps}}$$

$$(b) \quad 90 \text{ steps} \times \frac{8.0 \text{ in}}{1 \text{ step}} \times \frac{1 \text{ ft}}{12 \text{ in}} = \mathbf{60 \text{ ft}}$$

The step counts are exact, so the 8.0 in riser sets the precision: two significant figures. In metric, if it comes up:  $60 \text{ ft} \times \frac{1 \text{ m}}{3.28 \text{ ft}} \approx 18 \text{ m}$ .

$$(c) \quad t = \frac{90 \text{ steps}}{1.2 \text{ steps/s}} = \mathbf{75 \text{ s} = 1 \text{ min } 15 \text{ s}}$$

$$(d) \quad v = \frac{60 \text{ ft}}{75 \text{ s}} \times \frac{60 \text{ s}}{1 \text{ min}} = \mathbf{48 \text{ ft/min}}$$

One chain from the 60 ft in 75 s — there is no need to work out ft/s first and convert again. In metric, if it comes up:  $\frac{60 \text{ ft}}{75 \text{ s}} \times \frac{1 \text{ m}}{3.28 \text{ ft}} \approx 0.24 \text{ m/s}$ . Note this is the vertical speed; the distance walked along the stairs is longer, since each step also moves you forward.

7. **Challenge.** The conversion itself is right:

$$30 \frac{\text{mi}}{\text{h}} \times \frac{1.61 \text{ km}}{1 \text{ mi}} = 48.3 \text{ km/h} \approx \mathbf{48 \text{ km/h}}$$

What is wrong is the number of digits. The speedometer reads 30 mi/h — two significant figures — so the converted speed can carry at most two. The seven digits in 48.28032 came out of the calculator, not the measurement; they claim the car’s speed is known to a hundred-thousandth of a km/h. (The conversion factor is rounded too: 1 mi = 1.609344 km.)

Converting a measurement never makes it more precise — it only restates the same precision in a different unit.